GRISHMA BALGI

Brooklyn, NY | grishmabalgi@nyu.edu | (518) 955-1689
linkedin.com/in/grishma-balgi

Robotics Software & Navigation

EDUCATION

New York University | May 2027
M.S. Mechatronics and Robotics | GPA: 3.834/4.0

Veermata Jijabai Technological Institute | June 2025
B.Tech. Mechanical Engineering; Honors in Automobile Engineering | GPA: 9.04/10

TECHNICAL SKILLS

Programming: Python, C/C++, NumPy

Robotics: MuJoCo, kinematics, graph search; basic ROS exposure

Perception: RootSIFT, VLAD, topological navigation; OpenCV/MediaPipe project use

EXPERIENCE

Prema Lab, New York University | Sep. 2025 - Present
Graduate Research Assistant | Robotics fabrication, instrumentation and simulation

- Developing an 18-segment model of a fabricated TPU 95A tentacle in MuJoCo, representing passive bending with effective joint stiffness, damping and tendon elements.
- Iterating on geometry and passive response; investigating fluid-force effects while separating provisional model parameters from quantities requiring physical calibration.
- Fabricated a robotic-fin test rig and automated Silhouette Portrait 4 cutting through Python USB commands, generating coordinate-based paths for candidate fin geometries.
- Developed a motor-control interface and Arduino acquisition firmware to vary oscillation waveform, amplitude and frequency; logged load-cell readings alongside commanded and measured motor positions.

Schneider Electric | June 2024 - June 2025
Engineering Intern | Mechanical modelling and sensor systems

- Worked on reduced-order modelling of mechanical systems using Creo and Altair tools, preparing simulation-derived data and exploring models for circuit-breaker mechanisms.
- Built experience in sensor selection, vibration data acquisition and Python-based analysis for circuit-breaker condition monitoring.

SELECTED PROJECTS

Camera-Only Visual Navigation
Python, RootSIFT, VLAD, graph search | Course project | In progress

- Building a camera-only navigation pipeline using visual descriptors, a topological graph and Dijkstra search to navigate toward image-defined goals.
- Working on place recognition and geometric verification; evaluating extensions separately from the baseline rather than treating proposed methods as completed results.

UR10e Kinematics in MuJoCo
Python, NumPy, MuJoCo, numerical methods | Robotics coursework

- Implemented forward-position kinematics and numerical inverse-position solutions for a UR10e model within a supplied MuJoCo coursework framework.
- Explored velocity kinematics and pseudoinverse-based joint-velocity calculations, visualising robot motion in simulation.